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Influence of strategic approach to BPM on financial and non-financial performance

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Abstract

Purpose – The purpose of this paper is to examine the way strategic approach to business process management (BPM) impacts organizational performance, both its financial and non-financial aspects, using empirical data from Croatian firms. The impact of strategic approach to BPM on process performance measurement (PPM) is examined as well.

Design/methodology/approach – A questionnaire survey was conducted on a sample of 194 manufacturing and service firms in Croatia and propositions were tested using a structural equation model with SAS software.

Findings – The results suggest that PPM practice is positively related to strategic approach to BPM. The impact of PPM on non-financial performance has been found, as well as the impact of non-financial performance on financial performance, thus indicating an indirect influence of PPM on financial performance.

Originality/value – The paper extends the previous research that exclusively investigated impact of BPM to organizational performance. The authors extended results of previous research and found that strategic approach to BPM is an important push factor for implementation of PPM, and that PPM is an important link between BPM and improved organizational performance.

Keywords Croatia, Manufacturing industries, Service industries, Process management, Performance management, Financial performance, Non-financial performance, Business process management, Process performance measurement

Paper type Research paper

1. Introduction

Companies are striving for excellent business performance in order to survive in the global competitive environment. They are trying to achieve better results and higher profit margins by applying various emerging business tools and management philosophies. Business process management (BPM) is currently among the most important ones. Companies are emphasizing and continuously seeking to improve their performance management practices and performance measurement systems by providing them with a strategic focus and linking business strategy with bottom-line results.

Researchers engaged in performance measurement field are investigating how companies can achieve strategic objectives through the measurement of their performance. Although the link between strategy and performance management has already been widely researched, recent work (Kihn, 2010) emphasizes the need for



analysis of various types of performance outcomes. Future research should focus on strategic approach to performance measurement in general and process performance measurement (PPM) in particular.

Performance measurement has always been considered as a major instrument of performance management since it provides and integrates all information for making decisions related to the task of managing performance (Bititci *et al.*, 1997). However, since the discipline of BPM is still very young and rather undeveloped, the literature does not adequately address PPM, a newly recognized performance practice that could potentially offer significant gains.

The importance of PPM practices emerges from the distinctive nature of business processes. Such practices are essential for strategy execution and internal alignment. Moreover, they contribute to better control and management practices. While numerous papers have investigated performance management and measurement, studies examining the relationship between BPM and organizational performance are sparse. However, some recent findings (McCormack and Johnson, 2001; Škrinjar *et al.*, 2007, 2008) confirmed that business process orientation has a positive influence on organizational performance. Besides, Škrinjar *et al.* (2008) showed that the impact of BPM on financial performance is indirect through non-financial performance. Clearly, future research needs to place a strong focus on the strategic role of BPM and its process measurement dimension.

Notwithstanding the widespread understanding that strategic goals are achieved through performance and metrics driven management, the logic of performance measurement models needs to be modified and enriched (Taticchi *et al.*, 2010). They should clearly link strategic objectives and business processes on the one side with various performance measures on the other side. Therefore, the aim of the paper is to examine the impact of strategic approach to BPM on PPM, and, furthermore, to investigate how the latter influences both financial and non-financial performance measures.

The purpose of this paper is to explore if strategic approach to BPM has influence on financial and non-financial organizational performance. The influence of strategic approach to BPM on PPM is also explored. Survey data collected from a sample of Croatian firms has been used to examine possible impact.

The paper consists of five parts. After the introduction, the research model is explained and research propositions are developed. Next, research methodology is presented, including sample description and research instrument. Data analysis results and main research findings are provided in the fourth part. Finally, research results are discussed from both theoretical and practical standpoint, and directions for future research are outlined.

2. Theory and research propositions development

Research model

In-depth literature review and practical experience reveal that strategy and BPM represent holistic, intangible high-level concepts that have hardly been operationalized and would therefore be difficult to investigate directly. PPM, however, could play a positive and important mediating role in making the effects of strategy and BPM visible and measurable on the bottom-line. Although empirical evidence is lacking, we believe that managers can influence on their business results through PPM practice. By properly measuring organizational and individual efforts, managers send a clear

message about what is expected, which eventually mobilizes the workforce. In order to test our expectations and gain better understanding of possible relationships between strategic approach to BPM, PPM and performance outcomes, we have developed the research model as shown in Figure 1. The model is developed based on the following research propositions (RP), which will be briefly presented here. We presume that companies with a strategic approach to BPM actively deploy PPM in day-to-day business running (RP_1), which in turn has a positive impact on both non-financial performance (RP_2) and financial performance (RP_3). We also presume that PPM has an indirect positive influence on financial performance through non-financial performance (RP_4). In the following part of the paper, these presumptions are examined into detail regarding previous research.

In order to accomplish our main research goal and investigate the existence of direct and indirect influences between strategic approach to BPM, PPM practices, and both financial and non-financial performance measures, we proposed four research propositions. It was particularly important to address the dependent variable (performance outcomes) from a multidimensional perspective as financial and non-financial measures have different foci and can provide different results, which would eventually reveal interesting variances in performance practices. Definitions and research propositions relative to the variables and relationships present in the research model are discussed below.

Strategic approach to BPM and PPM

Organizations create and deliver value through intended actions that are governed by a chosen strategy. The strategy provides a framework for behaving and achieving organizational goals. It needs to be tightly integrated with enterprise business processes (Davenport, 1993; Spanyi, 2003, 2005; Ndede-Amadi, 2004; Neubauer, 2009; Brocke and Rosemann, 2010; Kohlbacher and Gruenwald, 2011). The strategic objectives are achieved through business processes, which in return generate value for the business. Strategy implementation and execution inevitably depend on processes and their interaction, are influenced by way process performance is measured, and rely on a deep understanding of how processes interface with customers (Harmon, 2007). Companies are literally defined by business processes, which potentially represent a significant source of competitive advantage.

BPM has become a discipline that can be used to improve the entire scope of organizational activities. As such, BPM practices represent a collection of methodologies, techniques, and tools supporting the analysis and improvement

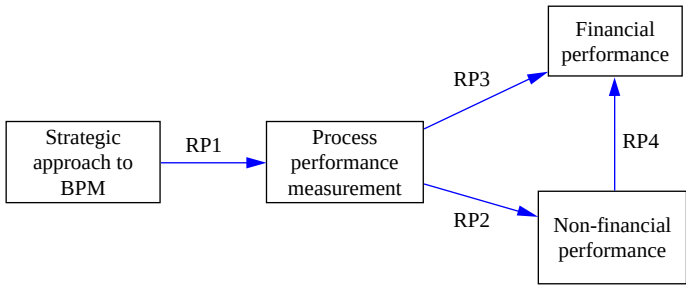


Figure 1.
The research model of the
process performance
measurement's impact on
organizational
performance

of business processes (Melao and Pidd, 2008). Using BPM framework managers try to align processes with the organization's strategic goals, design and implement process architectures, establish process measurement systems, and educate and organize managers to manage processes effectively (Burlton, 2003; Chaffey and Wood, 2005; Miers and Harmon, 2005). These challenges, among others, also include defining key performance indicators (KPIs) that align business process performance with business objectives.

Companies face a number of challenges in making BPM strategic. Most organizations, however, find that business process thinking helps to articulate strategic initiatives in plain language and with measurable results (Spanyi, 2005). Even common methodologies such as strategy maps have been developed (Kaplan and Norton, 2004) and they play an important role in linking strategy and process design. A superior process performance can be achieved by applying a strategic approach to BPM, which could drive existing bottom-line results and future destiny of an organization.

Kaplan and Norton (2000) were among the first researchers who emphasized that performance measurement systems need to be clearly linked to business strategy. Their request corresponds with the argument that performance measurement system aims to support the implementation and monitoring of strategic initiatives (Wouters and Sportel, 1995). Although the linkage and cause-and-effect relationship between strategy and process measurement is recognized in many BPM frameworks, empirical findings are still missing. With this in mind, we propose that a business strategy can be deployed and properly executed if the company employs a strategic approach to BPM and designs PPM practices to support it. In other words, a strategic approach to BPM will insure the implementation of process performance measures. Based on those findings we declare our first research proposition:

RP₁. Strategic approach to business process management positively influences implementation of process performance measurement.

Process performance measurement and organizational performance

Scholars have increasingly argued that business processes have an essential role in various aspects of doing business, including performance management, performance measurement and performance improvement. Performance measurement plays uniquely important roles in running an organization. It contributes to translating strategy into desired behaviors and results, communicating these expectations, monitoring progress, providing feedback, and motivating employees through performance-based rewards and sanctions (Chow and Van der Stede, 2006).

Performance measurement and ways to approach it are not clearly defined in the literature. Instead, the vast number of empirical studies produce an ambiguous picture of performance measurement (de Bruijn, 2002) and result in a great amount of disparate information. Due to inconsistent findings, the impact of performance measurement systems on organizational performance remains a much-debated question.

Performance measures and targets can be seen as a formulation of the strategic goals while the results achieved for these measures show how successful the company is in achieving the aforementioned strategic goals. By contrast, performance measurement system mostly comprises performance planning, work design, performance measures and reward system, with the objective of ensuring the

successful implementation of strategy (Lynch and Cross, 1995). It boosts performance improvement, which generally necessitates improving the processes. Moreover, process improvement and process performance issues are nowadays being widely addressed as performance measurement systems are introduced into organizations.

Research conducted by Pavlov and Bourne (2011) to evaluate the effect of performance measurement on performance reveals that the issue is poorly understood. The authors suggested that performance measurement affects organizational performance through affecting the dynamics of organizational processes. A similar argument was presented by Lynch and Cross (1995) who claim that in reality performance measures are seldom aligned or integrated with the business processes. Finally, Taticchi *et al.* (2010) suggested that a modification of performance measurement and management models is required. It is clearly necessary to develop new models, which will allow companies to identify relationships between processes their business is based on.

PPM is of a great importance because it helps to control, estimate and improve processes and organizations. It potentially leads to significant financial and non-financial improvements (e.g. increased revenue, cost reduction, cycle-time improvement, increased customer satisfaction, added value, employee satisfaction, better collaboration). Although the nature of relationship between PPM and organizational performance is not clear (Gonzales and Rubio, 2010) and is still not investigated properly, there are some indications of their positive relationship. The existence of positive relationship between PPM and organizational performance is even more likely to occur if PPM systems are supported from the top and aligned with strategic objectives.

Financial and non-financial measures of organizational performance

Performance measurement was traditionally strongly influenced by financial reporting, which resulted in the development of numerous financial measures. Most generally used financial measures include return on assets (ROA), return on equity (ROE), return on investment (ROI), profit margin, earnings per share, value per employee, etc. Even though they used to be very popular, these traditional financial measures are no longer seen as adequate means of exercising management control (Neely, 2007). Their weaknesses are well documented in the literature and include failing to convey strategies and priorities effectively within an organization (Najmi *et al.*, 2005).

A change in perspective took place in the mid-1980s. Performance measurement moved away from having a purely financial focus and organizations started to implement increasingly non-financial performance measures such as customer retention, customer satisfaction, employee turnover, and number of new products developed. It should, however, be emphasized that translating a strategy into specific objectives that guide operational actions requires both financial and non-financial measures. Rather than choosing only one type, financial and non-financial measures should be viewed as complementary to each other (Keegan *et al.*, 1989; Kaplan and Norton, 1992; Chow and Van der Stede, 2006; Kihn, 2010). The introduction of lagging and leading indicators into common performance framework was largely triggered by the work of Kaplan and Norton (1996).

Although a positive relationship between business process orientation and non-financial performance has been proposed and confirmed by several authors (Amaratunga *et al.*, 2001; Skrinjar *et al.*, 2007, 2008; Kumar *et al.*, 2008), there has been

little research into the specific influence of PPM, especially on the non-financial performance such as customer and/or employee satisfaction. However, various BPM models developed over the years clearly show that process measures and management systems play a more prominent role at the higher levels of process maturity. In addition, they relate to higher performance margins, both financial and non-financial (McCormack and Johnson, 2001; Raschke and Ingraham, 2010). Accordingly, we decided to propose the second research proposition:

RP₂. Process performance measurement practice positively influences non-financial performance.

A survey conducted by Altinkemer *et al.* (1998) showed that introducing BPM practices seems to be correlated with the productivity measure sales by employees, but their effect on other financial performance measures was not evident. Furthermore, Davis and Albright (2004) demonstrated that business units, which adopted the balanced scorecard model and its performance measurement framework significantly, outperformed other business units. Since the balanced scorecard model comprises internal process measures, there is reason to suggest that PPM has a direct impact on financial performance. The inference, however, should be more rigorously investigated. Based on the arguments presented above, we propose the third research proposition:

RP₃. Process performance measurement practice positively influences financial performance.

Hoque and James (2000) studied the use of non-financial measures proposed by various performance measurement frameworks and confirmed a strong positive relationship between such measures and financial performance. The findings suggest that there may be a certain time lag in the performance effects. Empirical analyses conducted by Škrinjar *et al.* (2007, 2008) produced similar findings: business process orientation has a strong, indirect impact on financial performance through non-financial performance. However, there are no findings in the literature that would specifically relate PPM with performance outcomes and positively confirm the nature of their relationship. Therefore, our fourth research proposition is as follows:

RP₄. Process performance measurement practice has an indirect positive influence on financial performance through non-financial performance.

3. Research methodology

Sample description

In order to test the research propositions, a questionnaire survey method was used to collect the responses of Croatian medium-sized and large-sized manufacturing and service firms. During June and July 2008, questionnaires were distributed to companies with more than 50 employees. The main source of data was the database of The Institute for Business Research. The questionnaire was sent randomly to one-half of the population, i.e. 1,800 companies. It was addressed to the CEOs or the chairpersons who were asked to fill in the questionnaire themselves or give it to a competent person within the organization.

The research group finally received 202 completed questionnaires, which represents a response rate of 11.2 percent. Several questionnaires, however, were not taken into

consideration due to missing data so the final sample consisted of 194 companies. Van der Stede *et al.* (2007) addressed the quality of survey research in management accounting, and found that surveyed articles reported an average response rate of 55 percent. This implies that the response rate in our survey is considerably low and that there may be differences between responding and non-responding firms. However, according to Dilman (1999), our response rate is comparable with similar research using mail surveys.

Sample data on companies examined (Table I) clearly indicate the predominance of medium-sized companies (67.18 percent). Furthermore, industry structure types were such that there were almost two manufacturing firms for each service firm in the sample. Finally, according to the annual turnover amount, the majority of companies (77.95 percent) had a turnover of less than 30 million EUR. In comparison to the population characteristics, our sample is somewhat biased towards manufacturing companies and/or companies with more than 500 employees with an annual turnover amount higher than 30 million EUR. χ^2 -test proved that these differences are also statistically significant at 1 percent level. Therefore, it is important to take into consideration these differences when evaluating the results of our research.

Research instrument

The research instrument consisted of three parts. The first two parts measured strategic approach to BPM and PPM practice. They had five items each and were designed to measure independent variables. The third part comprised organizational performance measures. We included three financial and 12 non-financial measurement items in order to test both financial and non-financial measures. Such an approach is based on the balanced Scorecard (Kaplan and Norton, 1992, 1996), and although somewhat different (we included external stakeholders perspective related not only to customers and employees but also to measuring the relationship with suppliers and process benchmarking perspective), it was designed to measure various facets of organizational performance.

The aforementioned constructs were previously developed through joint research activities of several scientists from the Faculty of Economics, University of Ljubljana and Faculty of Economics and Business, University of Zagreb. Our research activities

Firm characteristics	Sample characteristics		Population characteristics	χ^2	p-values
	Number of firms	Percentage	Percentage		
<i>I. Industry</i>					
Manufacturing	128	65.98	55.57	4.389	0.000*
Service	66	34.02	44.43		
<i>II. Employee number</i>					
< 250	131	67.18	81.18	16.362	0.000*
250-500	31	15.90	11.18		
> 500	33	16.92	7.47		
<i>III. Annual turnover amount</i>					
< 10 mil EUR	68	34.87	34.42	10.220	0.000*
10 mil EUR-30 mil EUR	84	43.08	53.29		
> 30 mil EUR	43	22.05	12.29		

Table I.
Comparative profile of
responding firms and
population characteristics

Note: *p-value < 0.01

were largely triggered by the initial work of McCormack and Johnson (2001). The multi-stage and multi-year process of construct development has been well documented elsewhere (Škrinjar *et al.*, 2007, 2010; Hernaus *et al.*, 2008; Kohlbacher, 2010). As the field is quite mature nowadays, the definitions of BPM dimensions do not display a significant level of diversity. However, in this paper a construct developed by Škrinjar *et al.* (2010) is used since the authors reviewed more than 15 different conceptualizations of might constitute process oriented elements. As the result of their research nine distinct dimensions were detected:

- (1) strategic view;
- (2) process definition and documentation;
- (3) process measurement and management;
- (4) process organizational culture;
- (5) people management;
- (6) market orientation;
- (7) supplier view;
- (8) process organizational culture; and
- (9) IT/IS support.

For the purpose of this research, we selected and/or synthesized several dimensions and their elements to develop a research instrument. All multi-item measures used were based on seven-point Likert scale from 1 (strongly disagree) to 7 (strongly agree) and are shown in more detail in the Table II.

Statistical methods

Data collected by the questionnaire survey method were analyzed in several steps and using various statistical methods. First, validity analysis was conducted and research instrument validity checked. Content validity has been ensured by using the items adapted from the literature, and by conducting the pilot study. Furthermore, convergent validity was tested using explanatory factor analysis in order to uncover the underlying structure of a relatively large set of variables, which were used under the a priori assumption that any indicator may be associated with any factor (Hair *et al.*, 2006). Discriminate validity was tested using confirmatory factor analysis in order to determine the degree to which measures of different latent variables are unique enough to be easily differentiated from other constructs (Costello and Osborne, 2005).

Second, reliability analysis was conducted using Cronbach's α coefficients that indicate internal consistency of the items used for calculating scales (Feldt and Kim, 2008).

Third, primary data were submitted to descriptive data analysis and non-parametric correlation analysis in order to check for negative or low correlations which could indicate some data validity problems (de Vaus, 2001).

Fourth, structural equations model fit was tested using the fit indices proposed by Hooper *et al.* (2008): χ^2 index, goodness-of-fit-statistics (GFI), adjusted goodness-of-fit-statistics (AGFI), normed-fit index (NFI), non-normed-fit index (NNFI), comparative-fit index (CFI) and root-mean-square-error (RMSEA).

Construct	Code	Item
<i>Business process management practices</i>		
Strategic approach to business process management	SBPM1	Top management is actively involved in process improvement efforts
	SBPM2	Business (sub)process goals are derived from and linked to the organization's strategy
	SBPM3	Business process improvement and redesign is frequently on the agenda of top management meetings
	SBPM4	Policy and strategy are communicated and cascaded throughout the organization
	SBPM5	Improvement plans for high-level processes exist and are driven by customer and operating strategies
Process performance measurement	PPM1	Process measures are defined and documented for each process
	PPM2	Process performance is measured in the organization
	PPM3	Performance targets are used for each process goals
	PPM4	Performance indicators are communicated within the organization on a regular basis
	PPM5	Performance results are used in setting improvement targets
<i>Financial performance</i>		
	FP1	Profitability of the firm increases faster compared to industry average
	FP2	Return on assets (ROA) of the firm is significantly higher than industry average
	FP3	Value added per employee is significantly higher than industry average
<i>Non-financial performance</i>		
External stakeholders management measures	ESM1	We retain existing clients and manage to attract new-ones
	ESM2	The number of customer complaints within the last period has increased strongly
	ESM3	Reputation of our company in eyes of the customers has improved
	ESM4	We consider our relations with suppliers to be excellent because we maintain genuine partnerships with them
	ESM5	There is a mutual trust between our company and our suppliers
	ESM6	Quality of our products is well above the industry average
Employee or HRM measures	HRM1	The net fluctuation of employees is very high within our company
	HRM2	Productivity of employees is much higher than industry average
	HRM3	Employees do not feel special commitment to the organization
	HRM4	Absenteeism is in our company (relative to competition) very high
Process measures	BPM1	Response time to customer complaints is well above industry average
	BPM2	Service error level (waste level) is much lower than our competitors'

Table II.
Research instrument
description

Finally, the structural equations model was used for statistical testing of the research propositions with the focus on the signs of the parameters, statistical significance of the parameters and the amount of variance of endogenous constructs accounted for by independent constructs.

4. Data analysis and research findings

Validity analysis

Before conducting multivariate analyses, research instrument validity was checked. Three dimensions of construct validity were investigated: content validity, convergent validity, and discriminant validity. Content validity was attained by adapting items from the literature and our previous research. In order to assess whether to expect an acceptable level of understanding, we piloted the questionnaire by interviewing ten managers. Certain changes were made as a result of several in-depth interviews.

Furthermore, explanatory factor analysis was conducted to test convergent validity. Statistical package SAS was used to run iterated principal axis factor analysis combined with varimax rotation for extracting six factors. Factor analysis results are displayed in Table III.

Regarding the cut-off values of factor loadings, we decided to use the moderate approach proposed by Costello and Osborne (2005) who recommended using loading cut-off value in the magnitude from 0.40 to 0.70. This moderate approach alleviates the concerns about the arbitrary nature of cut-off values, which could damage convergent

Item	Factor					
	1	2	3	4	5	6
SBPM1			0.8125			
SBPM2			0.6744			
SBPM3			0.5879			
SBPM4			0.5828			
SBPM5			0.5999			
PPM1	0.8234					
PPM2	0.7923					
PPM3	0.8674					
PPM4	0.7082					
PPM5	0.7997					
FP1				0.8845		
FP2				0.8652		
FP3				0.7200		
ESM1		0.6066				
ESM2		0.5854				
ESM3		0.7412				
ESM4		0.5689				
ESM5		0.7071				
ESM6		0.6763				
HRM1					0.3902	
HRM2					0.6438	
HRM3					0.6974	
HRM4					0.6414	
BPM1						0.5908
BPM2						0.6756

Table III.
Rotated factor matrix
for six factors

validity (Hair *et al.*, 2006). According to the established criterion, all but one of the measurement items from the questionnaire were to be retained. The only exception was HRM4, which loaded with the value of 0.3902. However, since the value for this item was close to the prescribed cut-off value and was used in the preliminary study by Škrinjar *et al.* (2010), we decided to include it in the analysis. All of the examined factors, except the sixth factor (BPM), consisted of three or more items, which is desirable. We used two items only for the measurement of the sixth factor as McCormack and Johnson (2001) originally proposed it.

The discriminate validity was also checked – the degree to which measures of different latent variables are unique enough to be easily differentiated from other constructs (de Vaus, 2001). For the purpose of this study, discriminate validity was assessed using confirmatory factor analysis. The factor loadings are shown together with *t*-values in Table IV. According to Costello and Osborne (2005), to be methodologically rigorous, all λ 's should be higher than the cut-off value of 0.50 and all of the *t*-values should exceed 1.96. Since all *t*-values were statistically significant at the level of 0.01, we may conclude that the observed loading paths were methodologically rigorous and relevant.

Reliability analysis

The questionnaire, originally developed in the Slovenian language, was translated into Croatian and its validity confirmed. Furthermore, reliability analysis was also conducted and Cronbach's α coefficients were calculated. In evaluating the scale reliability we were led by recommendations that internal consistency coefficients of

Item	Standardized factor loading	<i>t</i> -values	Cronbach's α
SBPM1	0.836	10.381	0.8827
SBPM2	0.780	9.776	
SBPM3	0.784	9.821	
SBPM4	0.691	1.0×10^{38}	
SBPM5	0.807	10.074	
PPM1	0.845	1.0×10^{38}	0.9348
PPM2	0.875	15.830	
PPM3	0.932	17.766	
PPM4	0.777	13.011	
PPM5	0.880	15.997	
FP1	0.939	23.762	0.9301
FP2	0.947	4.5×10^{08}	
FP3	0.825	17.126	
ESM1	0.791	13.858	0.8987
ESM2	0.685	11.058	
ESM3	0.883	1.0×10^{38}	
ESM4	0.718	11.857	
ESM5	0.790	13.837	
ESM6	0.805	14.299	0.8259
HRM1	0.520	1.0×10^{38}	
HRM2	0.882	7.507	
HRM3	0.841	7.374	
HRM4	0.760	7.041	
BPM1	0.957	4.8×10^{08}	0.7862
BPM2	0.675	9.005	

Table IV.
Standardized loading
estimates and *t*-values

0.70 or higher are considered to indicate adequate reliability (Feldt and Kim, 2008). As can be seen from the last column of Table IV, all Cronbach's α coefficients were above the cut-off value, which suggested that the item scales were internally consistent.

Primary data analysis

Following descriptive data analysis, a non-parametric correlation analysis was conducted in order to get a better research insight. Spearman's correlation coefficients have not revealed negative or very low (near-zero) correlations between items examined. Furthermore, all of them were statistically significant at the level of 0.01 (Table V).

Most coefficients showed that there was a moderate correlation between items representing strategic approach to BPM and PPM. The highest correlation coefficient values between individual items were as follows: PPM2 and SBPM5 ($\rho = 0.588$, $p < 0.01$); PPM3 and SBPM5 ($\rho = 0.533$, $p < 0.01$); PPM5 and SBPM5 ($\rho = 0.511$, $p < 0.01$); PPM5 and SBPM2 ($\rho = 0.505$, $p < 0.01$). The values determined empirically support the existence of the positive linkage between two constructs.

With respect to the relationship between PPM and dependent variables (both financial and non-financial performance measures), medium positive correlation was found in the case of non-financial performance, while most of financial performance indicators were only slightly related and not very significant. Stronger relationships were determined in the following dyads: BPM1 and PPM5 ($\rho = 0.445$, $p < 0.01$); ESM5 and PPM3 ($\rho = 0.438$, $p < 0.01$); ESM5 and PPM2 ($\rho = 0.429$, $p < 0.01$); HRM3 and PPM3 ($\rho = 0.418$, $p < 0.01$); BPM1 and PPM3 ($\rho = 0.413$, $p < 0.01$); ESM6 and PPM5 ($\rho = 0.413$, $p < 0.01$); ESM5 and PPM5 ($\rho = 0.410$, $p < 0.01$); and ESM6 and PPM2 ($\rho = 0.408$, $p < 0.01$). The results clearly emphasize that PPM practices are positively related to and have a significant influence on product or service quality. Additionally, PPM practices promote the establishment of strong mutual trust with suppliers.

Assessment of model fit

Assessment of model fit was conducted using the fit indices proposed by Hooper *et al.* (2008). The proposed conceptual model was developed using SAS module and yielded a χ^2 of 487.78 with 262 degrees of freedom (due to the sample size and number of variables used). Since χ^2 is usually sensitive to sample size, other indices were additionally used to assess the overall model fit (Table VI).

The correspondence between observed and hypothesized variance was assessed through GFI. Although it should be higher than 0.90, the GFI in our model was 0.835, which is very close to the recommended level. Adjusted goodness-of-fit index was 0.796, which is somewhat low but still acceptable. Furthermore, the values of NFI and NNFI were close to 0.90, which indicated a good fit (Bentler and Bonnet, 1980). CFI value was also acceptable although slightly below 0.95, while RMSEA indicated satisfactory 0.067 value (Steiger, 2007). In another words, we may conclude that our research model fulfilled the aforementioned rigorous methodological requirements.

Research propositions testing

Since the overall model exhibited a good fit, we were able to examine further the structural part of the model with the goal of estimating whether proposed theoretical relationships are supported in a specific research context. In the following analysis,

Table V.
Descriptive statistics and
correlation coefficients

	Mean	σ	SBPM1	SBPM2	SBPM3	SBPM4	SBPM5	PPM1	PPM2	PPM3	PPM4	PPM5	FP1	FP2	FP3	ESM1	ESM2	ESM3	ESM4	ESM5	ESM6	HRM1	HRM2	HRM3	HRM4	BPM1	BPM2
SBPM1	6.06	1.30	1																								
SBPM2	5.89	1.33	0.705*	1																							
SBPM3	5.30	1.56	0.552*	0.484*	1																						
SBPM4	4.37	1.55	0.544*	0.518*	0.503*	1																					
SBPM5	5.43	1.39	0.580*	0.481*	0.584*	0.537*	1																				
PPM1	4.90	1.67	0.302*	0.400*	0.391*	0.357*	0.473*	1																			
PPM2	5.09	1.57	0.420*	0.455*	0.487*	0.407*	0.588*	0.744*	1																		
PPM3	4.76	1.72	0.382*	0.485*	0.431*	0.435*	0.533*	0.768*	0.783*	1																	
PPM4	4.46	1.70	0.356*	0.453*	0.441*	0.533*	0.433*	0.631*	0.648*	0.723*	1																
PPM5	5.06	1.68	0.408*	0.505*	0.423*	0.441*	0.533*	0.433*	0.631*	0.648*	0.723*	1															
FP1	4.22	1.65	0.254*	0.259*	0.301*	0.276*	0.142*	0.150*	0.177*	0.178*	0.235*	0.159*	1														
FP2	4.21	1.54	0.260*	0.257*	0.273*	0.259*	0.148*	0.209*	0.248*	0.233*	0.229*	0.206*	0.871*	1													
FP3	4.44	1.58	0.273*	0.210*	0.297*	0.225*	0.171*	0.182*	0.218*	0.196*	0.210*	0.221*	0.747*	0.747*	1												
ESM1	5.49	1.38	0.408*	0.382*	0.377*	0.304*	0.349*	0.319*	0.363*	0.375*	0.313*	0.391*	0.497*	0.514*	0.616*	1											
ESM2	5.33	1.35	0.333*	0.242*	0.279*	0.224*	0.258*	0.265*	0.292*	0.328*	0.232*	0.364*	0.276*	0.253*	0.376*	0.532*	1										
ESM3	5.60	1.17	0.373*	0.376*	0.402*	0.329*	0.363*	0.290*	0.323*	0.315*	0.243*	0.387*	0.489*	0.477*	0.574*	0.702*	0.602*	1									
ESM4	5.57	1.28	0.370*	0.446*	0.368*	0.244*	0.315*	0.333*	0.380*	0.374*	0.284*	0.351*	0.383*	0.436*	0.443*	0.550*	0.479*	0.629*	1								
ESM5	5.69	1.15	0.468*	0.450*	0.426*	0.338*	0.375*	0.377*	0.429*	0.438*	0.361*	0.410*	0.328*	0.338*	0.346*	0.565*	0.621*	0.656*	0.697*	1							
ESM6	5.48	1.17	0.364*	0.407*	0.413*	0.265*	0.360*	0.383*	0.408*	0.373*	0.284*	0.413*	0.382*	0.394*	0.451*	0.621*	0.481*	0.664*	0.569*	0.597*	1						
HRM1	5.28	1.64	0.235*	0.262*	0.210*	0.262*	0.249*	0.185*	0.188*	0.247*	0.178*	0.292*	0.214*	0.257*	0.406*	0.356*	0.374*	0.456*	0.303*	0.398*	0.372*	1					
HRM2	4.82	1.47	0.329*	0.367*	0.428*	0.325*	0.306*	0.340*	0.328*	0.387*	0.287*	0.388*	0.556*	0.533*	0.665*	0.394*	0.450*	0.586*	0.528*	0.490*	0.563*	0.446*	1				
HRM3	5.16	1.41	0.412*	0.449*	0.353*	0.418*	0.340*	0.379*	0.397*	0.418*	0.334*	0.388*	0.436*	0.449*	0.501*	0.515*	0.367*	0.571*	0.540*	0.493*	0.488*	0.394*	0.701*	1			
HRM4	4.89	1.66	0.285*	0.335*	0.303*	0.332*	0.243*	0.196*	0.213*	0.260*	0.244*	0.319*	0.467*	0.433*	0.528*	0.479*	0.388*	0.531*	0.463*	0.417*	0.480*	0.425*	0.677*	0.625*	1		
BPM1	4.78	1.43	0.337*	0.381*	0.366*	0.292*	0.343*	0.345*	0.373*	0.413*	0.369*	0.445*	0.513*	0.541*	0.510*	0.539*	0.431*	0.625*	0.561*	0.560*	0.691*	0.399*	0.599*	0.520*	0.510*	1	
BPM2	4.51	1.37	0.316*	0.361*	0.318*	0.306*	0.312*	0.282*	0.265*	0.372*	0.372*	0.306*	0.420*	0.397*	0.349*	0.362*	0.292*	0.357*	0.485*	0.421*	0.386*	0.253*	0.432*	0.380*	0.402*	0.615*	1

Note: * p-value < 0.01

Table VI.
Fit indices for the
research model

Fitness indicator	Model estimated	Explanations
Chi-square (χ^2)	487.77	χ^2 is not significant
Degrees of freedom (df)	262	
p-value	0.000	
χ^2/df	1.861	Very good, close to 2
GFI	0.835	Good, close to 0.9
AGFI	0.796	Good
NFI	0.874	Good, close to 0.9
NNFI	0.928	Very good result
CFI	0.938	Very good result
RMSEA	0.067	<0.07, a good result
90% confidence interval of RMSEA	(0.058-0.076)	Upper limit < 0.08, a good result

our attention was focused on the signs of the parameters, statistical significance of the parameters (measured by t -value) and the amount of variance of endogenous constructs that is accounted for by independent constructs (measured by the multiple correlation coefficient squared – R^2).

The results of the path analysis are shown in Figure 2. It is obvious that the path coefficient for the RP_3 has a low statistical significance. As path coefficient (0.309) was lower than the cut-off value of 0.5, the proposed research proposition had to be rejected. Other research propositions (RP_1 , RP_2 and RP_4) were supported with the path coefficients higher than the cut-off value and were statistically significant at the 1 percent level.

The RP_1 , suggesting that strategic approach to BPM is positively related to PPM practice, was accepted. The standard solution of path coefficient estimate from strategic approach to BPM to PPM was 0.696 with the t -value of 8.031, which indicated the existence of a positive effect at 1 percent significance level. In addition, R^2 -value

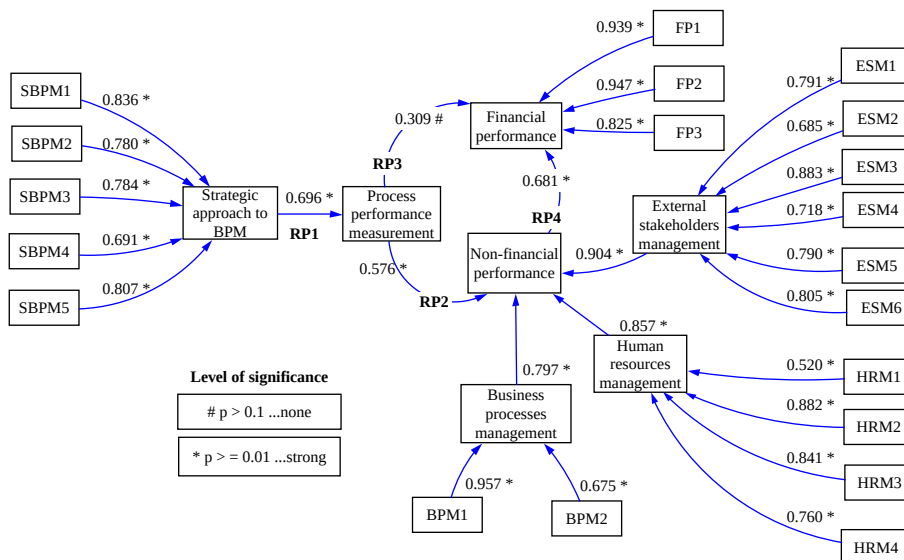


Figure 2.
Path diagram with path
coefficients estimates and
their significance levels

was 0.484 indicating that 48.4 percent of variations in PPM could be explained by variations in strategic approach to BPM.

The RP_2 , stating that PPM practice positively influences on non-financial performance, yielded the standard solution of path coefficient estimate from PPM to non-financial performance of 0.576 with the t -value of 6.966. The results clearly show the positive nature of the relationship at 1 percent significance level. In addition, R^2 -value was 0.331 indicating that 33.1 percent of variations in non-financial performance could be explained by variations in PPM.

As previously mentioned, the RP_3 , suggesting that PPM directly influences financial performance, was rejected. Namely, the standard solution of path coefficient estimate from PPM to financial performance was 0.309 with the t -value of 1.394, which indicated that the positive effect was present between the constructs. However, R^2 -value was 0.096 indicating that only 9.6 percent of variations in financial performance could be explained by variations in PPM, which eventually led to the rejection of the third research proposition.

Finally, the RP_4 , proposing that PPM practice has an indirect influence on financial performance through the impact of the non-financial performance, was also accepted. The standard solution of path coefficient estimate from non-financial to financial performance was 0.681 with the t -value of 9.372, which supported the existence of a positive effect between them at 1 percent significance level. In addition, R^2 -value was 0.464 indicating that a respectable amount of variance in financial performance could be explained by variance in PPM.

5. Discussion and conclusion

Relations to previous findings

An examination of the literature is undertaken to review studies that report about the relationship between a strategic approach to BPM and organizational performance through PPM practices. A systematic review of more than 40 papers, from 1992 to 2011 describing relevant issues, was conducted. The majority of findings have been presented into previous sections, while here the most important findings are revised.

The empirical study conducted by Neubauer (2009) analyzed current BPM practice, trends and strategic plans of companies. The survey showed that only a small minority of the companies have completely associated their processes with the strategy. According to the results, one of the most important fields that companies need to address is the association of the business strategy with the business processes and the systematical integration of BPM into long-term business objectives. On the other hand, the research results (de Waal *et al.*, 2009) show that organizations that have fully completed a strategic performance management system implementation gain more financial and non-financial benefits.

According to Kohlbacher (2010) the present review of literature has identified a clear lack of quantitative studies exploring the effects of process orientation. The analysis made by Milanović Glavan (2011) shows that today there is an extensive literature on BPM, business process orientation and business processes, an extensive literature on performance measurement, performance measures and performance measurement systems, but less literature, even the lack of literature on process performance measurement systems (PPMS). Only 12 articles that have the term PPMS in the title were found in this search. This concept was first introduced by Kueng (2000), but it started to

be more attractive only since 2010. It can be concluded that today's performance measurement systems are still not focused upon business processes.

BPM needs to be incorporated into the firm's strategy, otherwise it is doubtful that the high expectations onto BPM at the outset will ever materialise. On the other hand, PPM is a necessity for a modern, process-oriented organization. As such, it requires a new measurement approach. Finding the right way to systematically measure the impact of business processes and their success becomes a requirement. However, it still needs to be determined to what extent PPM as a concept influences on business results and contributes to them. Motivated by the scarcity of theoretical and research findings in this area, our research was designed to explore whether PPM plays a critical role in determining organizational performance, and to determine how strategic importance and managerial guidelines for BPM and PPM relate to financial and non-financial outcomes. Therefore, it is important to empirically research and test possible relationships between strategic approach to BPM, PPM and performance outcomes.

Furthermore, our research has also empirically supported some earlier assumptions regarding the leading role of non-financial indicators and the lagging role of financial indicators. McCormack and Johnson (2001) recognized the link between BPM and organizational performance, and Škrinjar *et al.* (2007, 2008) verified this link in transition economy with more detailed specification of organizational performance that included both financial and non-financial performance indicators. Our research extends those two previous papers. Based on empirical data we supported research propositions regarding impact of strategic approach to BPM on the implementation of PPM. In addition, we investigated and empirically confirmed the proposition that PPM is an important link between BPM and improved organizational performance.

Concluding remarks

This paper provides a very useful source for academics and companies since it investigates connections between BPM and business strategy. Three out of four research propositions have been supported with empirical data, which suggests that PPM practices provide better business results when aligned with business strategy and strategic importance of BPM. Namely, research findings revealed that the existence of business process goals and policies in general, and improvement goals and plans in particular, foster PPM practices. Not only that improvement effort inside organizations is evaluated through PPM practices; but also performance results are often used for setting updated improvement targets.

Besides, the paper sheds a new light on the relationship between PPM and organizational performance. The direct and indirect influence of strategic approach to BPM on financial and non-financial performance was investigated through PPM practices. A path analysis was conducted using structural equation modeling technique and research propositions were tested on the survey data collected from 194 Croatian firms.

The existence of positive and direct relationship between PPM and non-financial performance has also been supported by empirical data. Organizations, which have developed performance targets for each process goal and used them for setting further improvement actions reported a response time to customer complaints well above industry average. In addition, PPM practices also helped organizations to build a

relationship of mutual trust with their suppliers, which eventually enabled them to offer products and services of higher quality than their competitors.

Furthermore, the very interesting and somewhat neglected role of non-financial measures has been addressed and supported by empirical data. The study has shown that PPM has an indirect impact on financial performance measures through non-financial measures. Profitability was strongly predicted by process measures, customer satisfaction and employee measures, while ROA and value added per employee were mostly predicted by employee measures. These findings indicate that non-financial measures demonstrate performance improvement over a short-term period, i.e. they are leading indicators. Financial measures, on the other hand, have long-term effects and can be seen as lagging indicators. In other words, managers should not only concentrate on financial data. Monitoring various non-financial performance indicators will considerably contribute to making good decisions in a real-time manner.

Research limitations

Our research, however, has several limitations. First, the research sample could be more extensive in order to get more generalized results. It is hard to conclude from a single-country study whether the tested research propositions are valid in different countries and other research contexts. It is desirable to conduct similar studies but on a larger and international sample of companies in order to be able to make valid generalized conclusions.

Second, a cross-sectional design was used in our study. Although such a research design is ideal for conducting descriptive analyses, we used it to test a causal model of the influence of strategic approach to BPM on financial and non-financial performance through PPM practices. Consequently, our research had to deal with the internal validity problem in establishing cause-and-effect relationship without a time dimension. In order to resolve the issue of internal validity, future research activities should include a cohort analysis as a series of repeated cross-sectional studies. Such a design could be used to describe aggregate change in financial and non-financial performance over time and identify the extent to which this change is attributable to the influence of strategic approach to BPM through PPM practices. In addition, a longitudinal research design would enable us to test the causality determined in this study. It could also be effectively conducted through applying a multiple case study approach.

Implications

Our research findings are of high importance for both research and managerial practice. Regarding possible academic implications, given results have empirically established the importance of PPM practice and confirmed its mediating role between strategic approach to BPM and organizational performance. Not only that PPM practice has been supported with empirical data as an important element of BPM concept, but has also been suggested as an important tool for strategy implementation and execution. Moreover, it has become clear that not only performance measurement systems in general, but also PPM systems in particular, should be linked to business strategy.

The literature dealing with the relationship between BPM and organizational performance is still rather scarce. Empirical studies of the link between PPM and its financial and non-financial outcomes are even rarer. This study is, therefore, a good starting point for future research about PPM systems and BPM frameworks.

Research results should be also used as a starting point for development of case studies investigating the link between BPM and PPM. These case studies should also explore how PPM is used for measuring financial and nonfinancial performance. It would be important also to find examples from different industries and different company sizes.

Business practices of companies that implement strategic approach to BPM and PPM into its day-to-day business practice are also more transparent, which is important prerequisite for decreasing the level of corruption, especially in transition countries. In addition, employees that are knowledgeable on how their working output will be estimated regarding PPM would also feel less uncertainty at their workplaces.

The managerial implications of the research are at least twofold. First, the research findings regarding the role of PPM practices will inevitably encourage companies to examine their current performance measurement practices in more detail and/or motivate them to introduce new systems oriented towards measurement of their business processes. It is to be expected that those systems will be aligned with the company's business strategy. Such an active managerial approach towards PPM could improve both financial and non-financial performance of companies and make them more competitive on the global market. Second, the tested importance of non-financial measures, which are more informative and less time-constrained than financial ones, will induce managers to pay more attention to indicators such as customer satisfaction, relationship with suppliers, employee commitment and satisfaction, as well as quality and process measures. Furthermore, non-financial measures are easily applicable and can be used by individual companies to predict their financial performance beforehand. Improved understanding and clearly demonstrated financial and non-financial benefits of implementing and practicing PPM will result in a wider application of such systems in everyday business practice, which will eventually lead to their refinement and further development. We look forward both to future investigations and to increased application of PPM in practice, as these will provide significant gains for companies worldwide.

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